

Fall 2024 CSCI 605 Assignment#5:

Implementing a Binary Search Tree

Objective:

In this assignment, you will implement a Binary Search Tree (BST) where each node stores a key-value pair. The key will represent a name (as a string), and the value will represent a phone number (as a string or number). You will then implement a search functionality that allows you to find the phone number associated with a given name.

Instructions:

1. Class Design:

- Create a class `TreeNode` to represent a node in the binary search tree. Each node should store:
 - A `key` (the name).
 - A `value` (the phone number).
 - References to the left and right children of the node.
- Create a class `BinarySearchTree` that represents the binary search tree. The tree should be initialized as empty and should have methods to:
 - Insert a new node with a name and phone number.
 - Search for a phone number given a name (the key).
 - Optional: Delete a node by name from the tree (extra credit).

2. Implementation Details:

- **Inserting into the Tree:**
 - Implement a method `insert(name, phone_number)` in the `BinarySearchTree` class that inserts a new `TreeNode` in the correct position according to the binary search tree property:
 - Names that are lexicographically smaller go to the left subtree.
 - Names that are lexicographically larger go to the right subtree.
- **Searching in the Tree:**
 - Implement a method `search(name)` that returns the phone number associated with the given name. If the name does not exist in the tree, the method should return `None` or an appropriate message.
- **Traversals:**
 - Implement an `inorder_traversal()` method to print all names and their associated phone numbers in sorted order.

3. Extra Credit (Optional):

- Implement a method `delete(name)` to remove a node from the tree based on the given name.
- Consider edge cases like deleting a node with no children, one child, or two children.

Example Usage:

Your `BinarySearchTree` class should work as shown in the example below. This is just an example, your test cases should have much more records:

```
bst = BinarySearchTree()
bst.insert("Alice", "555-1234")
bst.insert("Bob", "555-2345")
bst.insert("Charlie", "555-3456")

# Searching for phone numbers
print(bst.search("Alice"))    # Output: 555-1234
print(bst.search("David"))    # Output: None (or "Name not found")

# In-order Traversal
bst.inorder_traversal()
# Output:
# Alice: 555-1234
# Bob: 555-2345
# Charlie: 555-3456
```

Grading Criteria:

- Correct implementation of `TreeNode` and `BinarySearchTree` classes.
- Correct functionality of `insert()` and `search()` methods.
- Working traversal method (`inorder_traversal()`).
- Optional extra credit for implementing `delete()` method.

Submission Requirements:

Submit your implementation as a Python file with the class definitions and methods. Ensure your code is well-documented with comments explaining the logic behind each part of the implementation. You should also include a few test cases to demonstrate that your tree functions correctly.

- Due Date: 10/14/2024

- **Do not use ChatGPT or other AI tools to create your code.**
- Create a Github project that includes the following:
 - Name of your repository: CS605_Fall24_<Firstname>_<Lastname>
 - Within your repository, create project: Assignment_5
 - Within your project:
 - Source code
 - A readme file describes how to use your code
 - A test document describes the test cases and what you observe in your tests
 - A screenshot of how your code is successfully executed
- Submission:
 - On Canvas, link to your Github repo/projects
 - Make sure you have invited me to your github repo/projects (see Additional Notes)

Additional Notes:

For submission of github links, you need to either make your github repo public, or add me as a collaborator. Here is an instruction on adding people as collaborators.

<https://docs.github.com/en/account-and-profile/setting-up-and-managing-your-personal-account-on-github/managing-access-to-your-personal-repositories/inviting-collaborators-to-a-personal-repository>

You can find me through using my email: bshen2@csueastbay.edu or my github account name: boshen-csuchico.